

Anritsu VNA with S-parameter Measurement

The VectorStar vector network analyser (VNA) from Anritsu breaks the radio-frequency (RF) barrier by spanning from 70 kHz to 70 GHz in a single connector. For applications ranging from 70 kHz to 2.5 GHz, a mixer-based receiver with bridges for directional devices is used. Above 2.5 GHz, a harmonic-sampling receiver is incorporated, with traditional couplers for directional devices. VectorStar features a completely new VNA platform that uses an innovative architecture.

Using this equipment, you can perform S-parameter measurements of RF, microwave and millimetre-wave devices. It provides a powerful measurement tool for performance analysis of devices ranging from transistors in an on-wafer environment to communication systems in commercial or defence applications. (For details, visit www.us.anritsu.com)



VTI High-density Signal Switching System

For test, product validation and systems engineers who are responsible to automate the functional testing of complex electronics systems, here's a high-density multifunction signal switching sub-system. Up to 576 channels of voltage or temperature can be measured in a single-rack 1U mainframe, with additional capabilities including DC to 26.5GHz RF switching, isolated analogue outputs, arbitrary waveform generation, digital input/output (I/O) and edge detection. The LXI Class A communications interface (as claimed by VTI) can be used to create a highly synchronised, distributed measurement system that performs data logging, monitoring and control.

The equipment provides measurement support for all thermocouple types, RTDs and thermistors with built-in cold junction compensation. It has a powerful inbuilt Web-based graphical user interface (GUI) that provides virtual monitoring and control of all switches and instruments without the need for third-party software. An integrated 6.5-digit digital multimeter increases the test throughput. (For details, visit www.vtiinstruments.com)



fully automated X-ray inspection systems have just started gaining popularity, especially in the food and drugs industry, and surface-mount technology (SMT) production.

At the recent *Apex Expo* on T&M technologies for SMT production, some companies such as Test Research Inc (TRI) demonstrated products with X-ray inspection technology that can see through printed circuit boards (PCBs) and hard-to-inspect components to check their connections to the PCB. Such test systems enable in-line inspection of high-density components such as flip chips (controlled collapsed chip connections), ball grid arrays (BGAs) and micro-components down to 0201 and 01005. These use X-ray penetration to capture images through several cameras positioned at different angles. Software is then used to analyse the slice images and calculate the different slice heights to inspect for PCB defects.

Another technology that has been around for over three years, but has suddenly gained more popularity, is 3-D imaging. According to a recent study by ITM Marketing, "The primary end-user applications for automated optical inspection (AOI) systems are solder-paste inspection and process optimisation. Increasingly, users want 3-D imaging in their systems." Basically, in these systems, light is focused on the

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